

Social Media Sentiment As A Predictor Of U.S. Stock Market Returns

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1 Background & Purpose

On January 28th, 2021, GameStop stock (\$GME) began trading at over \$500 per share in pre-market, attracting significant media attention, inflicting large losses on some institutional investors, and triggering trading halts on several platforms. The short squeeze was preceded by months of activity on the `r/wallstreetbets` subreddit, where many retail investors discuss investment ideas and strategies. However, the language used in that community diverges sharply from standard English, to the point that it functions almost as a distinct financial dialect.

This project investigates whether market returns can be effectively predicted by machine learning models trained on `r/wallstreetbets` text and historical stock price data. Social media information flow is fast; if the subreddit contains meaningful price signals and suitable models can extract them in real time, then the approach could support highly lucrative trading strategies. In particular, if such a technique had existed and been deployed during the 2021 meme-stock episode, it might have captured a portion of the extraordinary price movements observed at that time.

2 Current State Of The Art

Many state-of-the-art approaches rely on some combination of: keyword counts or keyword volume; transformer-based sentiment models trained on standard or formal text (e.g., VADER, FinBERT); predicting movements of a broad index rather than individual securities; and estimating returns over relatively long periods (for example, weeks or months rather than next-day horizons). These approaches are largely between 5 and 15 years old, and they were tuned to linguistic and market regimes that differ from the 2020–2021 meme-stock environment.

Social media discourse evolves rapidly, and the financial slang and norms used by communities such as `r/wallstreetbets` diverge over time from the language captured in older models. Market behavior is also non-stationary. As institutional traders adopt earlier “state-of-the-art” methods, any simple, exploitable structure they detect typically becomes arbitrated away and incorporated into prices. Thus, both the linguistic and market contexts motivating earlier work differ meaningfully from those considered here.

2.1 Text Interpretation

Prior work on financial text analysis has largely focused on more formal sources and general-purpose sentiment:

- Loughran-McDonald Master Dictionary (Loughran & McDonald, 2011)
- VADER (Hutto & Gilbert, 2014)
- Sentiment analysis on Twitter data integrating TextBlob and deep learning models (Aljedaani et al., 2022)
- FinBERT (Araci, 2019)

These tools are valuable benchmarks, but they are not tailored to the slang, sarcasm, and meme-heavy style prevalent on `r/wallstreetbets`, which may limit their ability to extract fine-grained sentiment from this domain.

2.2 Market Prediction From Text

Several studies have investigated the relationship between social media sentiment and financial markets:

- Twitter mood predicts the stock market (Bollen, 2011)
- Results likely invalid per *Shy of the Character Limit: “Twitter Mood Predicts the Stock Market” Revisited* (Lachanski & Pav, 2017)
- Correlating S&P 500 Stocks with Twitter Data (Mao et al., 2012)
- The Effects of Twitter Sentiment on Stock Price Returns (Ranco et al., 2015)

Most of this work focuses on Twitter rather than Reddit, uses general sentiment tools rather than domain-specific lexicons, and typically targets broad index movements. In contrast, the present study emphasizes ticker-level predictions and Reddit-specific discourse.

A key methodological component of this project is the use of short-horizon time-series modeling. Temporal convolutional networks (TCNs) are a recent architecture for sequence forecasting and often outperform recurrent models on time-series tasks. *Temporal Convolutional Networks Applied to Energy-Related Time Series Forecasting* (Lara-Benitez et al., 2020) reports substantial gains over traditional recurrent architectures, making TCNs a strong candidate for short-duration stock return prediction as well.

3 Our Approach

The study compares several modeling strategies for stock return prediction: models that use only price history, models that use only sentiment features, and models that combine sentiment and price history. Within each strategy, multiple sentiment analysis methods and time-series modeling architectures are evaluated.

3.1 Data Collection & Preprocessing

The dataset consists of approximately 53,000 posts from the `r/wallstreetbets` subreddit, spanning late September 2020 through mid-August 2021. For each post, the title and body were concatenated into a single text field, and usernames, email addresses, mentions, and URLs were removed. Posts that became empty after cleaning were dropped.

To link posts to tradable securities, lists of tickers from the S&P 500, NASDAQ, and Dow Jones Industrial Average were used to build association tables between tickers and company names. Related instruments were grouped (for example, NDAQ with QQQ and QQQM; S&P 500 with SP500, SPY, SPX, VONG), and frequently mentioned non-index tickers such as GME, AMC, and BB were manually added. Tickers that coincide with common English words were treated carefully: some (A, BRO, COO, L) were ignored outright, while others (ARE, HAS, IT, NOW, etc.) were only recognized when written in all caps. Regular expressions, combined with these rules and association tables, were then used to identify, for each post, the set of tickers referenced.

For each identified ticker, daily adjusted close prices were retrieved from Yahoo Finance from seven trading days before the ticker’s first mention in the dataset to seven trading days after its last mention. Adjusted close prices account for stock splits and dividend payouts, allowing for consistent return calculations.

Additional corporate event information (earnings dates, ex-dividend dates, stock split dates, and dividend payout dates) was also collected from Yahoo Finance. These events can induce significant, often abrupt price changes that are not driven by Reddit sentiment. To explore the impact of such events on model performance, the dataset was prepared in multiple variants that differ in how aggressively these dates are filtered.

3.2 Sentiment Analysis

Two sentiment tools were applied to each cleaned post: FinBERT and a custom `r/wallstreetbets`-specific sentiment model. FinBERT is a transformer-based model trained on financial texts, which provides domain-aware sentiment estimates for earnings reports and other formal

financial documents. The custom tool is deliberately simple: it scans the post text for keywords from predefined bullish and bearish wordlists, and then assigns a continuous score by adding one point per bullish term and subtracting one point per bearish term.

Both sentiment scores were standardized using Z-scores. Under this transformation, the average sentiment across all posts becomes 0, a sentiment one standard deviation more bullish than average is mapped to 1.0, and a sentiment one standard deviation more bearish than average is mapped to -1.0 . This scaling places FinBERT and custom sentiment on a comparable numeric footing and stabilizes model training.

3.3 Model Architectures

For the price-only approach, the following architectures were considered: Ridge regression, Random Forest regression, Linear Regression, a univariate LSTM, and a CNN-LSTM hybrid that attempts to capture local and longer-range temporal patterns in prices.

For the sentiment-only approach, the standardized scores from both sentiment tools served as input features to Ridge regression and XGBoost models. Finally, the combined sentiment-and-price-history approach used the two sentiment features along with the past seven trading days of prices as inputs to Ridge, XGBoost, LightGBM, and TCN models. In each case, models were evaluated across multiple prediction horizons and event-filtering configurations.

3.4 Grid Search Over Return Days & Event Filtering

Each model was trained and tested under a grid of two key conditions: prediction horizon and event-filtering strictness. The prediction horizon determines the target return. Denoting the trading day of a post as $p + 0$ and the next trading day as $p + 1$, the models were tasked with predicting returns for horizons from $p + 1$ through $p + 7$. The return for horizon $p + k$ was defined as the adjusted closing price at $p + k$ divided by the adjusted closing price at $p + 0$.

Event-filtering strictness controls how observations around major corporate events are handled. Several levels were explored: no filtering; on-date (removing only posts on event dates); adjacent-date (removing posts within one day of an event); very near date (removing posts within three days); and near date (removing posts within seven days). These levels were combined with the different horizons to create a systematic grid of training conditions.

3.5 Input Data Aggregation

To reduce label redundancy and noise, inputs were aggregated at the ticker-day level. For each ticker on each calendar day, all posts mentioning that ticker were grouped and the associated sentiment scores were averaged. The models thus consumed an “average daily sentiment” for each ticker instead of a set of many posts with identical price labels but potentially conflicting sentiment. Model performance improved markedly after this aggregation step, highlighting the importance of smoothing noisy social media signals before prediction.

3.6 Train/Test Data Ordering

All observations were sorted chronologically and then split into training and test sets, preserving temporal order. This prevents information from the future from leaking into the past (for example, via random shuffling) and ensures that evaluation reflects a realistic forward-looking forecasting scenario.

3.7 Our Contribution To The State Of The Art

The approach makes several contributions relative to existing work. First, it focuses on a high-volume, near-real-time data source—`r/wallstreetbets`—that has received less systematic attention in the context of short-horizon equity return prediction. Second, it incorporates a custom sentiment model tailored to the subreddit’s slang and tone, thereby attempting to bridge the gap between meme-driven discourse and existing sentiment analysis tools such as FinBERT.

Third, the analysis is conducted at the individual-ticker level, rather than solely at the index level, which allows for a more granular assessment of how retail sentiment might relate to specific securities. Finally, the work combines sentiment features and historical prices as input to TCNs, and no prior study was found that uses this particular combination of Reddit-based sentiment, event-aware filtering, and TCNs for equity return prediction. Taken together, these elements represent a novel application of existing machine learning techniques to a new problem domain.

4 Results

The best-performing model, measured by root mean squared error (RMSE) of return predictions, was a TCN that used both sentiment and price time-series inputs. This model achieved an RMSE of approximately 0.023 on its target horizon. However, across all con-

figurations, no model was able to predict the sign of the return with accuracy meaningfully above 50%, indicating that directional predictions were effectively indistinguishable from random guessing.

Two robust patterns emerged across models and settings. First, performance was consistently best for the $p + 1$ horizon; prediction errors generally increased as the horizon moved further into the future, which is consistent with the intuition that uncertainty compounds over time. Second, with respect to event-date filtering, many models improved as filtering became more aggressive, up to a point. Under the strictest filtering level, performance deteriorated for several models, likely due to the substantial reduction in training data. Interestingly, models targeting $p + 1$ returns tended to perform best with strict event-date filtering, suggesting that near-term returns around major events are particularly noisy.

These interpretations are supported by the plots and heatmaps referenced in Figures 1 through 5.



Figure 1: Performance comparison for models predicting price solely from previous prices.

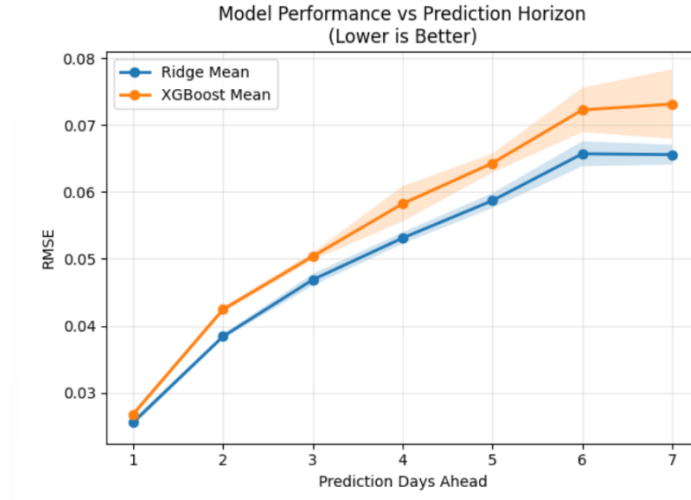


Figure 2: Performance comparison for models predicting return from sentiment only, $p + 1$ through $p + 7$.

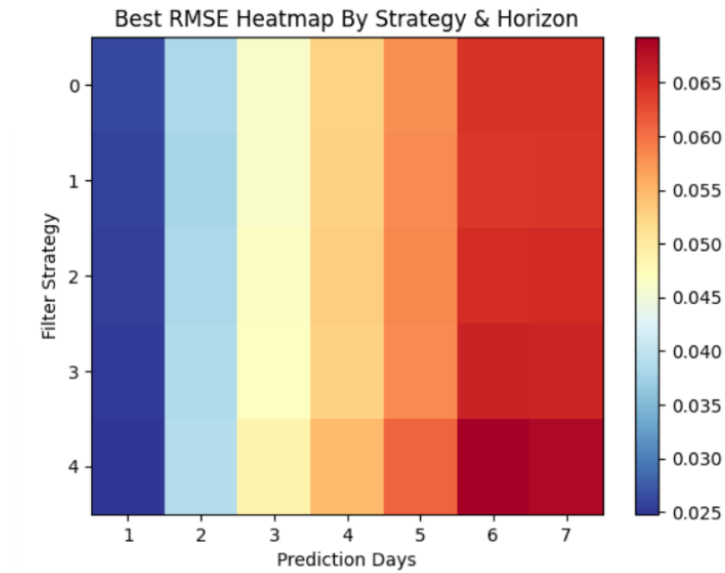


Figure 3: Performance heatmap across filtering (0 meaning no filtering, 4 meaning strictest filtering) for the XGBoost sentiment-only return estimator model.

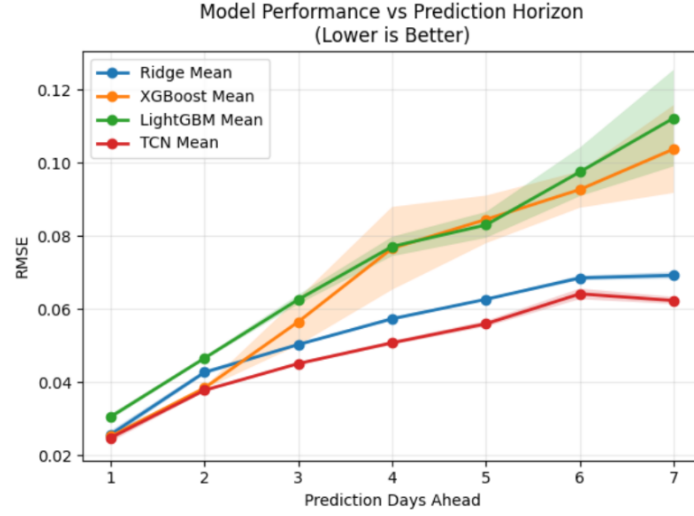


Figure 4: Performance comparison for models predicting return from sentiment *and* historical price data, $p + 1$ through $p + 7$.

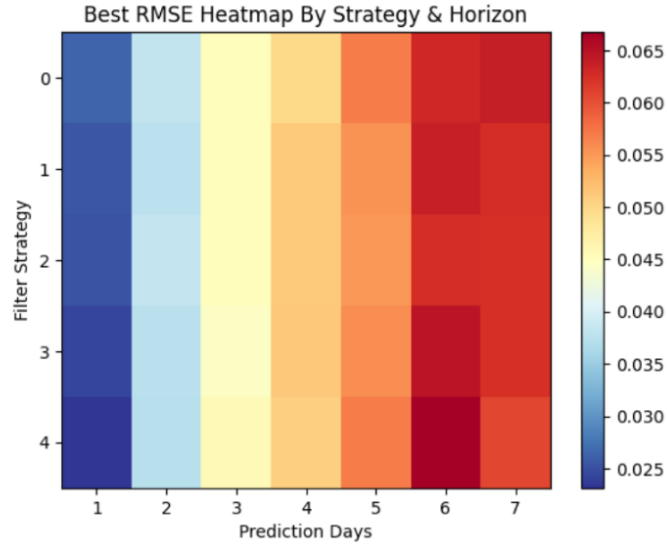


Figure 5: Performance heatmap across filtering (0 meaning no filtering, 4 meaning strictest filtering) for the TCN sentiment+price return estimator model.

5 Conclusion & Future Research

None of the models evaluated here performed well enough to support profitable trading based solely on `r/wallstreetbets` data. Nonetheless, the results validate several components of the modeling pipeline. Aggregating sentiment at the ticker-day level and filtering around major event dates both improved performance, and the $p + 1$ horizon consistently emerged as the most tractable prediction target. The TCN’s superior RMSE relative to other models is consistent with the broader literature on TCNs as a strong architecture for time-series forecasting.

Two features of the dataset likely constrained performance. First, the sample covers less than a full year of `r/wallstreetbets` activity. After cleaning, event-date filtering, and sentiment aggregation, the initial 53,000 posts collapse to just over 5,000 usable ticker-day samples across the combined train and test sets. Second, those samples span dozens of tickers with heterogeneous volatility and cross-correlations, which introduces additional noise and makes it more difficult for models to learn stable, ticker-agnostic patterns.

Despite these limitations, a TCN that attains an RMSE of roughly 2.31% across many tickers is promising. Substantially expanding the dataset—for example, by including a longer history of `r/wallstreetbets` posts, incorporating trading volume and order book features, or integrating cross-platform sentiment—could provide richer signals for the same modeling framework. Future work should therefore focus on scaling data collection, refining domain-specific sentiment tools, and exploring more expressive temporal architectures within this event-aware, sentiment-plus-price modeling paradigm.